Keeping the Organization in the Loop as a General Concept for Human-Centered AI: The Example of Medical Imaging

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Abstract

This study emanates from work on human-centered AI and the claim of "keeping the organization in the loop". A previous study suggests a systematic framework of organizational practices in the context of predictive maintenance, and identified four cycles: using AI, customizing AI, original task handling with support of AI, and dealing with contextual changes. Since we assume that these findings can be generalized for other kinds of applications of Machine Learning (ML), we contrast the management activities that support the four cycles and their interplay with a widely different domain: the usage of AI for radiology. Our literature analysis reveals a series of overlaps with the existing framework, but also results in the need for extensions, such as holistic consideration of workflows or supervision and quality assurance.

1. Introduction

Various approaches attempt to capture how the roles between humans and artificial intelligence (AI) – particularly machine learning (ML) – are changing with respect to autonomy and distribution of decision making. In this paper, we focus on the contribution of human-centered AI (HCAI) approaches [1]–[3] to this discussion. HCAI argues generally for employing a socio-technical perspective (for instance [1]), and for keeping the human in the loop [4]. We understand HCAI as an approach that seeks to recognize and to employ human capabilities, and to further develop them when accomplishing goals by using AI. Although organizational alignment is considered an indispensable component of socio-technical systems [5], HCAI approaches do not systematically consider the organization.

We conceive organizations as a multidimensional and paradoxical phenomenon, as an entity that is integrated into complex external environments and performs there as an actor; as structures and processes – such as hierarchies – of responsibilities, and competencies, all of which are subject to their own logic and

rules [6]. The actual characteristics of these levels de-pend on the development of everyday organizational practices. Decision-making by individual members of the organization – even when supported by ML – de-pend as much on diverse and conflicting interests as on collective action. All this is not only strategically orchestrated by management in organizations, it is also negotiated and processed within organizational prac-tices on a daily basis. Therefore, in the following we use the term "organizational practices" because such practices are crucial for how an organization is trans-lated into reality.

To take organizational practices into account sys-tematically, we follow Herrmann and Pfeiffer's [7] concept of “keeping the organization in the loop” (KOITL). We have chosen this recently published ap-proach because it fills the meso-level gap between keeping individual humans and keeping the society in the loop [4]. However, their framework relies empiri-cally on case studies in the single field of predictive maintenance (PM) in industrial manufacturing, beg-ging the question of its generalizability.

Therefore, we investigate the question (RQ1) of what changes to the KOITL framework must be made if applied to domains that are fundamentally different from PM. To answer RQ1, we chose ML applications for medical-image detection in the field of radiology. This field is comparable with PM in that it also in-volves well-established and complex workflows. It differs from PM in that the purpose of radiology is to prepare medical treatment for people, where the effi-cient use of resources is of secondary consideration ra-ther than the primary goal as in PM. PM is a strategy for detecting the condition of an equipment to identify maintenance needs to avoid faults and malfunctions, but “not to do it too often from the viewpoints of reli-ability and cost” [8, p. 135]. For PM, large sensor data generated in industrial plants can be used to train ML to detect undiscovered correlations or critical anoma-lies [9]. By contrast, ethical issues and privacy con-cerns are of higher relevance in the field of radiology. Radiology seeks data sharing between institutions to

URI: https://hdl.handle.net/10125/103278
978-0-9981331-6-4
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improve its quality and usage for machine learning [10], while PM prefers to keep most of the data within the company. Due to the relevance of applying huge datasets, we refer to ML as the most relevant version of AI in this paper.

Although radiology is only one exemplary field, we suggest that it provides enough contrast to industrial PM so that we can go one step further when discussing the generalizability of the keeping the organization in the loop framework. In what follows we analyze literature on the interplay between HCAI and the organizational perspective (Section 2). Since many case studies about implementing ML in the field of radiology are already documented in the literature, this paper focuses on a systematic literature review to identify the relevant facets of organizational practices of ML usage in radiology (Section 3), followed by the description of our findings (Section 4). The discussion compares the findings in both areas (Section 5), and summarizes our recommended modifications to the KOITL framework.

2. Organizational Challenges: Neglected in Human-Centered AI Research

In the engineering and social science research on human-AI interaction, organizational practices are either entirely overlooked or discussed only tangentially to topics like productivity, biases in personnel management, or profitability. Some human-centered AI research occasionally points to organizational factors in the socio-technical context [1] or to organizational factors such as the initial integration of AI with situated work practices and organizational processes [11]. Others suggest that explainable AI (XAI) has to be complemented with network-based trust [12]. Organizational practices are supposed to be needed to implement principles for responsible AI [13] or to achieve reliability and security [14].

The sociology of technology has gotten past a dichotomization of human action versus technical operation, replacing it with the more differentiated framework of “distributed agency” [15][16]. However, these approaches focus on sociological analysis rather than organizational design. The organization is addressed more as a static environmental condition than as a dynamic set of actors and practices. The interaction between humans and AI is often reduced to a few aspects or even just one, e.g. how deeply the outcome of AI affects humans [17], how individuals are connected to a technical system, or how organizational measures might achieve certain productivity gains [18]. Some authors discuss only the distribution of moral decision-making as in the case of autonomous driving [19].

Overlooked are the multiple layers between the micro level of human-computer interaction (HCI) and the macro level of societal regulation [4]. Between these lies the meso level of organizational practices, in which the way of using technology is concretized and processed as well as changed and developed on a daily basis. However, the assumption is that it is a once-off-process of integration, and the organizational practices remain mere context of an ongoing process of main-taining the integration [20]. Rarely does current re-search address that AI can be the subject of disputes [21] or of questions of power [22]. What is currently needed is a systematic overview of how AI and organ-izational practices have to be intertwined as a prereq-uisite and continuous context of human-centered AI. This gap may be practically justified by pursuing a clear focus of research. However, it has a systematic background, as conducting empirical research that ad-dresses the organizational context might be more chal-lenging than conducting HCI laboratory studies [23].

We argue that a technical concept for HCAI must be accompanied by a specification of the organiza-tional practices that precondition its success. For ex-ample, interventions in automated processes [24], [25] or vetoing an AI outcome [26] require an organiza-tional context that explicitly promotes or at least al-lows these actions. Similarly, the technical approaches that seek to provide XAI [27] cannot succeed if the or-ganizational workflows do not provide time and op-portunities to use XAI features. Concepts of how to activate and develop the complementary strengths of humans and AI [1] or to avoid oppressive ML systems [28] require appropriate organizational practices, for example sequences that have humans make their deci-sions [29] before considering proposals provided by ML. Within the Information Systems discourse, the need for managerial activities – such as communi-cating, leading, coordinating, and controlling [30] – or for organizational practices that overcome the per-ceived limitations of AI in radiology [31] is acknowl-edged, and the question of how roles will change is considered [32].

These approaches are complemented by Herrmann and Pfeiffer [7], who provide an approach that proposes a systematic and detailed framework of considering organizational practices that support HCAI. They conducted case studies in the context of predictive maintenance (PM) – considered to be one of the most important fields of ML applications for in-dustry 4.0 [9]. Their KOITL framework identifies four types of managerial activities (Table 1) and, within these, 12 sub-activities. Managerial coordination (Ta-ble 1, field 1) is necessary for determining how ML-based PM notifications are handled and – if applicable – used for maintenance. This is closely related to the

leadership and HR task of making people competent to use AI and of determining the roles of those inter-

practices, its original development depended on observations from the field of industrial predictive maintenance alone.

acting with AI (Table 1, field 2). The quality of managerial decisions also depends on coordination with an external environment (Table 1, field 3) that comprises all stakeholders and related organizational units that are not part of the plant that is subject to the PM application. Similarly, ongoing changes have to be considered (Table 1, field 4) with respect to the contextual factors that influence the performance of a PM-system. Despite the analytical separation shown in Table 1, the four types of activities are highly interrelated. They all interact with and shape the use of AI outcomes for task-handling and for evolving the AI system. We focus on the managerial activities shown in Table 1 because they allow a direct comparison with organizational measures proposed for using ML within radiology.

Table 1. Types of Managerial Activities Supporting Organizational Practices (following [7], Fig. 2).

1. Coordination of AI-related tasks	**2. Leadership an**
a. Defining workflows and new tasks of using AI for original tasks b. Defining the ecology of relevant roles c. Reflection of interdependencies between contextual factors and AI	a. Aligning role b. Employing a perience and c

3. Coordinating with the external world
a. Regulation of access from the outside
b. Identifying and observing various and varying external influences to be addressed
c. Regulating the sharing of risks and benefits

Herrmann and Pfeiffer [7] characterize the metaphor of keeping the organization in the loop by four cycles and their interplay, which all have to be intertwined with the managerial activities described in Table 1. The first two cycles refer to dealing with AI (using and assessing AI output; customization of the AI system). The third cycle refers to the original tasks that are intended to be supported by AI. Taking changes and contextual factors into account for evolving an AI system is emphasized by the fourth cycle. All four cir-cles are intertwined. The KOITL framework [7] has an important limitation. Although it aspires to provide a theoretical view of relevant facets of organizational

3. Systematic Literature Review

While Section 2 was based on a narrative review [33], for the literature on organizational aspects of us-ing ML-based medical imaging in the field of radiol-ogy we conducted a systematic literature review (SLR) that allows for reproducibility and borrows from Brocke et al. [33]. To develop an overview on apply-ing ML for radiology, we referred to Wang and Sum-mers [10] and Liang et al. [34].

The systematic literature review focuses on the question, what statements or topics regarding organi-zational or managerial decisions on measures in the field of radiology can be found? We expect that group-ing similar statements and comparing the set of these groups with the framework provided by Herrmann and Pfeiffer [7] will point out new requirements for reflect-ing and adapting the original framework.

To delimit the area of interest we started with the search term >“medical imaging” AND “radiology” AND “machine learning” AND “human-centered” AND “organizational”< (August 31, 2022) and in-cluded publications from the last 10 years. We used Google Scholar to include a wide scope of interdisci-plinary research (144 hits), Web of Science (0), Sco-pus (15), IEEE Explore (0) and ACM digital library (7). The inclusion of the term "human-centered" was the reason that this search yielded so few hits.We used Scopus to see if broadening the search would yield more relevant results. To this end, we searched litera-ture that also included AI or ML as keywords and also resulted in hits when either "medical imaging" or "ra-diology" or both were mentioned. This test did not yield any additional relevant literature after applying the exclusion criteria noted below.When we used terms such as "management" or "organization" in ad-dition to "organizational," the search results were pre-dominantly unsuitable for our purposes because these terms were semantically ambiguous (e.g., "the organi-zation of this paper" or "information management sys-tem"). When we replaced “organizational” by “socio-technical”, the search results proved too narrow (27 hits with Google Scholar). However, we found one ad-ditional relevant source [35] that was not identified with the first search term.

In all, we identified 21 relevant literature sources using the following selection criteria:

- long papers written in English published in jour- nals or conference proceedings (except one highly relevant thesis),

- papers that dealt explicitly with for reporting [41] or making measures and practices in context of radiology work – but not those that merely mention that the

cient [42]. These tasks are related to different interests of many participants and stakeholders. A wide variety of tasks, departments, professions, employees at various hierarchical organizational level is relevant,

- papers that consider the ML, not just its development.

We identified 19 further papers in the course of the narrative review, by applying backward and forward research, and by consulting an expert in the field of ML for medical imaging who pointed to the activities of the European Commission [36] and the US Food & Drug Administration [37]. Note that a literature search for publications dealing generally with the relation between ML and organizational practices would have revealed much more literature than the search focused on radiology alone. However, this limitation is justified for the purposes of this paper, which seeks to contrast two application domains on the concrete level of their typical workflows.

In order to match the content of the identified contributions with the KOITL-framework, we searched the 40 papers for statements on organizational or management-related challenges or measures. We consider it an advantage that the contexts of this papers are quite diverse, as we aim for an exploratory search that re-veals a variety of different aspects. The contributions were systematically subjected to a content analysis ac-cording to Mayring [38]. Identified statements were first grouped by searching for semantically similar statements or subclasses that fit the four management activities summarized in Table 1. In a second step, we proceeded without predetermined coding and explora-tively found new categories. This approach resulted in a total of seven categories, three of which are not in-cluded in Table 1 and include 15 new subcategories. In the next section, we present these categories and in-dicate in parentheses after each subheading how many papers contained information contributing to the cate-gory (one paper can be relevant for several categories).

and their roles with respect to medical imaging have to be clarified [43].

Appropriate, seamless workflow integration [39] and reconfiguration of processes [44]–[46] is a key challenge for the success of ML applications [47] be-cause of the highly situated nature of activities in clin-ical environments. ML has to contribute to a ‘‘best’’ procedure for a given clinical circumstance, including factors such as diagnostic power, radiation exposure, etc. [48]. ML applications should be well integrated into the underlying technical infrastructure, covering image archiving and communication systems [49] and procedures of providing and curating the data needed to train or retrain and evaluate ML models [48]. Thus, the entire life cycle of ML implementation needs sup-port [50].

4.2.Leadership: Promoting Complementary Strength and Customization of AI(16)

Employees make countless experience-based deci-sions in their daily work [51] that are affected by ML. Some statements from studies emphasize the strength of ML [52], e.g. how algorithms can detect details that would be neglected by humans [49] [53]. However, most papers argue that the relative strengths of human radiologists and ML systems are complementary [10], [54]. ML systems "… are mainly viewed or treated as an aid or helper in a secondary position" [10, p. 947]. Consequently, radiologists are expected to remain in-volved and engaged [55] so as to improve quality and efficiency [53], e.g. to guarantee forward-looking re-sponsibility [35]. Human strengths lie in adapting to practice patterns, maintaining relationships, communi-cating findings appropriately, and dealing with incom-plete information, e.g. of scans [42] [55] [47].

4. Findings

4.1.Coordination of a Variety of Original Tasks and Workflows (13)

Applying ML in radiology not only covers selected tasks such as the analysis of CT or MR images but a whole workflow [39] of activities such as medical im-

To employ and to support human strengths, cer-tain

age registration, brain function or activity analysis, content-based image retrieval systems, medical image segmentation, text analysis of radiology reports [10], or detection of similarities between images to clarify a diagnosis [40]. Thus, a broad range of technical sup-port is relevant, such as natural language processing

tion to highly complex constellations or refining the search of images related to the case at hand [41] [40]. As a further example, organizational measures can specify that the human should first clarify his or her own expectations before being presented with an AI output [29] [48] [56].

Organizational structurers should allow human end-users to innovate new technologies [57], e.g. by providing feedback to the ML system to enhance its possibilities by processing new data [48]. According to Cai et al. [40], when using an interaction mode for refining ML output, users experienced less effort and

more benefits, and methods of experimenting with the system's outcomes led to higher trust.

4.3.Coordination with the External World: Net-

of the identified literature. Algorithms, quality meth-ods, performance measures, monitoring, feedback, and accountability have to be combined [48]. Quality re-quirements can be grounded in ethical criteria [61]

working (9)

For the continuous improvement of ML-based medical image detection, studies reported several institutions cooperating in a networked effort [50], building strategic partnerships [57], and developing trust, e.g. between hospitals [58]. This concerns image sharing networks, mutual access to verified-case datasets, and exchange of experience and criteria for optimization and standardization [53], as it takes place within the emerging field of radiomics [59]. It pursues extracting useful imaging features from radiological data [48], [60], detecting relations between image-based findings and genetics, optimizing treatment and decisions about biopsies [59], and supporting a broader scope of clinical decision making [39]. Thus, radiomics helps to integrate the knowledge of many radiologists. However, data exchange is basically affected by increasing privacy regulations and ethical considerations [43].

4.4.Handling Contextual Factors and Changes (3)

Establishing AI in the practice of radiology – and its impact once established – are matters of continuous change in relation to advances in technology. Thus, a constant re-evaluation of relevant criteria and values is

[36] [62] covering aspects such as informed consent,

required [48]. These dynamics make continuous training necessary and therefore are dependent also on leadership tasks (4.2). The success of ML in regard to diagnostic performance, nursing workflows, and patient experiences depends on several contextual factors [47]. For example, monitoring the evolving legal context and adapting to regulations is a continuous organizational task [43].

4.5.Investing Additional Resources (5)

In addition to selecting appropriate ML algorithms, AI initiatives require a justification for the investment of significant resources such as hardware, software, and the human expertise (both computational and medical) [48] into which ML has to be efficiently integrated [58]. High-quality ground-truth data [39] comprising a substantial number of verified cases are a basic resource for ML training [53]. Providing these data is costly and requires regulated exchanges between institutions, expert involvement, and substantial time [49].

4.6.Ensuring Supervision and Quality (17)

The quality of ML-based results and the supervision of these systems is an important issue throughout most

privacy and data protection, ownership and data qual-ity, objectivity, human agency and oversight, technical robustness and safety, transparency, diversity, fair-ness, accountability, etc.

A particular quality challenge is the deployment of ML out of highly controlled environments into widespread use [55], or from one context to another [49] [42]. The question arises whether the modifica-tion of validated models by individual institutions or by empowered end-users should be permitted or whether the ML code should be frozen [55]. Modifi-cation needs testing and organizational regulations for quality assurance and possibly the involvement of the software developers' expertise [48]. There are con-cerns that the data used to train ML systems might be a source of uncertainty [56] [39].

Not all types of ML image-analysis applications are suitable to be subject to human monitoring [48]. A review process for ML results must be organized in-volving experts or institutions such as the American College of Radiology Data Science Institute [39] or the US Food & Drug Administration [37]. Radiolo-gists must receive training for supervisory tasks [41]. Further organizational challenges are the compensa-tion for patients affected by low-quality outputs [52] in compliance with evolving laws [43]. Obviously, dealing with quality challenges involves non-tech-nical, organizational measures [36] such as policies for accountability [63], governance frameworks [64], guidelines [58], or checklists [65].

4.7.Considering the Subjective Factor and Com-munication between Stakeholders (11)

Even if quality problems with the technical systems can be solved, this does not necessarily mean that the people concerned have confidence in technically de-rived analyses. If people trust human decisions more than machine decisions par tout, the benefit of ma-chine learning for radiology will be limited [10]. If people are provided with explanations about an ML result, one has to consider that their different profes-sional and educational backgrounds will affect their perception of such explanations. Therefore, different people who are part of the social and organizational environment in which an ML application is embedded might differ in their degree of trust in the application’s outputs [66]. Currently, the general population does not support a fully independent use of such systems without involving a radiologist [67], and physicians

are concerned about their own reputation and relevance [68]. Radiologists are still unsure about their future role and their responsibility for AI outcomes in relation to unsolved ethical and legal issues [69, p. 10]. They have low confidence in the results such systems

perspectives [50] [62]. A common "language" has to be established between multiple stakeholders includ-ing physicians, scientists, technologists, industry lead-ers, and patients [48].

generate [70]. Medical students are also skeptical [71]. Thus, the subjective attitudes – as well as the goals [62] – of different stakeholders need to be addressed at the beginning of AI implementation [11].

Exchanging knowledge in clinical practice requires much communication and mutual understanding, as well as the integration of multiple stakeholder

Table 2: The extension of management activities

1. Coordination of AI-re- lated tasks a. Defining workflows and new tasks of using AI **within workflows of** original tasks (4.1) b. Defining the ecology of relevant roles (4.1) c. Reflection of interde- pendencies between contextual factors and AI (4.4.) **d. Checking for integra- tion of all supports (4.1)** **e. Adding resources (4.5) f. Preparing the interplay between quality as- sessment and adapta-tion of AI (4.6)** **g. Supporting exchange between stakeholders' perspectives (4.7)**	3. Coordinating with the external world a. Regulation of access from the outside (4.3) b. Identifying and observing various and varying ex- ternal influences to be addressed (4.3) **c.** Regulating the sharing of risks and benefits, **e.g. of exchanging data and technology (4.4)** **d. Building strategic part- nerships (4.3)** **e. Support and regulation of cross-institutional exchange of underlying data (4.3)**
5. Supervision and quality assurance (4.6) a. Organizing supervision b. Organizing tests when new algorithms are de- ployed c. Regulating the modification of ML algorithms d. Quality assurance for c Organizing the involvement of external expertise	4. Dealing wit changes a. Assigning m various types b. Tracing caus problems (4. **c.** Deciding abo **e.g. of exch ogy (4.4)** d. Organizing n level long-ter changes **e. Observing a changes (4.**

5. Discussion

In what follows we discuss each of the categories of the findings and decide where an extension of management activities (Table 1) is reasonable. These ex-tensions are indicated in Table 2 in boldface. In paren-theses, we additionally refer to the subsection that mainly supports the subcategories of Table 2.

The distinction proposed by Herrmann und Pfeiffer [7] between AI usage and the original tasks AI is designed to support may also reasonably be applied to radiology. However, ML-based image detection influences a much broader scope of tasks than PM. The entire image-detection workflow context (see 4.1) is affected (Table 2, 1a). Thus, the suggested [7] inter-play between coordinative management activities with AI use for original tasks should include checking for integration of all kind of possible support with AI/ML (Table 2, 1d) and seamlessly integrating AI into work-flows (see 4.1). This integration specifically includes also organizational practices such that generated data can be exchanged, compared, and processed to serve as a basis for mutual knowledge and for advancing ML within the entire ML lifecycle. Furthermore, the prac-tice of radiology, similarly to PM, is structured by the various roles to which tasks, rights and duties have been assigned (Table 2, 1b, 2a). For example, not only image detection but also the task of image-scanning is an integral part of the relevant workflow, and manage-ment has to decide whether nurses are authorized to judge the quality of a scan [47].

The relevance of complementary strengths (see 4.2) in the field of clinical ML support is similar to what is found in the PM context (Table 2, 2b). Rele-vant as well is the promotion of human strengths and its interplay in refining AI results and in contributing to AI evolution through continuous customization, e.g. by retraining the ML models. The current proposals of how human strengths can be involved to help custom-ize AI (see 4.2) are mostly technical. However, the HCAI discourse (see Section 2) points to the involve-ment of organizational practices in making customiza-tion possible in the course of quality assessment (Ta-ble 2, 1f) and in preparing the users accordingly (Table 2, 2d). We assume that in the case of the PM-related framework, preparing the user for contributing to ML adaptation is also of certain relevance, but it is not ex-plicitly mentioned as a management activity in Herrmann and Pfeiffer [7]. It was, however, implicitly addressed in that leadership is considered relevant. We

suggest listing these preparation tasks (Table 2, 2c to 2e), since they are explicitly discussed as being relevant for medical practitioners [11] .

During the development and use of ML-applications, multiple communication exchanges take place between stakeholders with widely different backgrounds and experience. Thus, building strategic partnerships is relevant (Table 2, 3d). Additionally, medical ML is characterized by a higher relevance of cross-institutional exchange of ML-relatable data (Table 2, 3e) as demonstrated by the example of radiomics as a developing field (see 4.3). While in the field of radiology the exchange of data has to be regulated primarily under ethical and privacy aspects, for PM it is of interest to avoid access to competition-relevant data from

ology (see 4.7), it is highly relevant for the coordina-tion between dealing with AI and the original tasks (such as diagnosing and treating) that communication with patients and relatives and involving them into de-cision making is included (Table 2, 1g, 2e). This sub-jective factor seems to be of higher relevance in the case of radiology than with PM.

When expanding Table 1 to Table 2, it becomes clear that the five types of management activities must be closely intertwined to induce sustainable organiza-tional practices in support of HCAI. These practices also determine the extent to which management ac-tions are effectively implemented in dealing with AI and the use of AI results for the original tasks (e.g., repair work or medical treatment). In both domains,

the outside.

The deliberate consideration of external factors and their continuous change (4.4) appears to be rele-vant in both areas of medical imaging and PM. Both, for example, have to be aware of and prepared for con-tinuous technical change and challenges (Table 2, 4a, 4b, 4d). While PM is more focused on the economic

the notion of ML systems as autonomous agents [72]

context, e.g., availability or price of spare parts, radi-ology must additionally deal with societal changes, such as ethical and political discourses (Table 2, 4e). Of particular interest to radiology is how risks and benefits are shared when data or ML algorithms are exchanged (Table 2, extensions of 3c and 4c).

does not play a dominant role, yet it is recognized that ML technology is dynamically evolving, as described as a basic feature of AI [30]. Other, more general over-views of relevant organizational measures in the con-text of AI that address management challenges [30], principles of human-AI collaboration [73], or mixed human-AI initiative [74] do not provide the level of detail found in the field of radiology and PM in terms of considerations of management activities and their interplay with the use and adaptation of AI in the con-text of existing workflows.

The implementation of ML-based image recognition can be cost-intensive and requires many additional resources (see 4.5). This aspect has to be added to the management activities of coordinating AI related tasks (Table 2, 1e).

As ML-based predictions can have a direct impact on patient well-being, the reliability and validation of the quality of ML results (see 4.6) is of high relevance. This is reflected in the discussion about different methods of quality assurance and about organizational measures that should take place. Because of the high relevance of quality assurance and supervision, this aspect is newly inserted in Table 2 as an extension of the PM-related findings (Table 2, 5). It comprises mana-

6. Conclusion

gerial activities such as organizing supervision, the testing and regulation of modifications, and involving external expertise. These measures are intertwined with leadership (preparing staff for the shortcomings of AI and for supervisory tasks; Table 2, 2d) and coordination (preparing the interplay between quality assessment and adaptation; Table 2, 1f).

To be considered also is that different stakeholders (see 4.7) – all of whom differently affected – influence how ML systems and their outcomes are subjectively experienced. Staff have to be prepared to deal with this subjective factor (Table 1, 2e). Within radi-

The analysis of the two widely different domains points to the conclusion that keeping the organization in the loop (KOITL) is of general relevance for hu-man-centered AI. The analyzed papers hint at various organizational measures being relevant. For radiology, there is no theoretical framework for analyzing the interplay between ML and organizational practices. To fill this gap and with respect to the research question (RQ1), we extended the framework of Herrmann and Pfeiffer [7] by the extensions boldfaced in Table 2. Those managerial activities that had already been identified (see Table 1) also proved just as relevant in the context of radiology.

We suggest that Table 2 provides a theoretical ba-sis for how the interplay between AI usage and customization, the support of original tasks by AI, and dealing with contextual factors and ongoing changes can be supported by managerial activities. Supervision and quality assurance are added as an additional type of management activity that might prove of general relevance for implementing HCAI since they point to human supervisory tasks. The KOITL framework, which includes these extended management activities, can help in the creation of checklists for practitioners.

However, such checklists would need to be adapted to each specific domain context [23].

A limitation of our results is that they are based on a literature review rather than additional case stud-ies. However, there are already a number of case stud-ies reported in the literature found. Nevertheless, cer-tain aspects may be usefully explored in new case studies, e.g., whether adaptation and alignment can be promoted organizationally or whether this is prevented by regulatory contexts. Furthermore, it would be inter-esting to explore implementation in a third domain, e.g., AI applications for human resource management, in which bias avoidance and fairness are more central issues. Alternatively, the details found in this work could be compared to more general frameworks of management challenges as increasingly discussed in the Information Systems context [30].

7. Acknowledgement

This work was supported by the Humaine project (Human Centered AI Network), funded (02L19C200) by the German Federal Ministry of Education and Re-search (BMBF), and by the Deutsche Forschungsge-meinschaft (DFG, German Research Foundation) – project number 442171541. We thank Axel Mosig for hints to complete the literature review.

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